
MODEL SELECTION AND VALIDATION REPORT (fishing_kde_feve model)

OBSERVATIONS:1739

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1_with_anisotropy

MODEL SUMMARY (fishing_kde_feve model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: student(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	0.03	0.10
log_sum_fishing	-0.04	0.02
sbt_mean	0.01	0.06
sbt_sd	-0.04	0.04
sst_sd	0.07	0.03
sic_mean	-0.03	0.05
log_chla_mean	-0.01	0.03
depth	0.03	0.03

Dispersion parameter: 0.66

Spatiotemporal AR1 correlation (rho): 0.82

Matérn anisotropic range (spatiotemporal): 86.2 to 977.6 at 37 deg.

Spatiotemporal marginal AR1 SD: 0.42

REML criterion at convergence: 2389.486

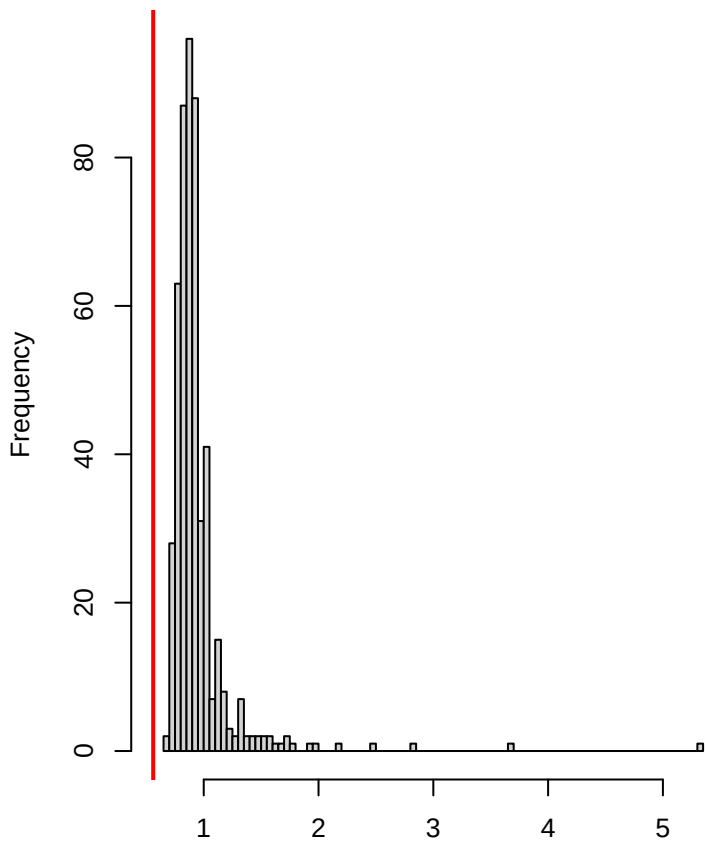
See ?tidy.sdmTMB to extract these values as a data frame.

See ?plot_anisotropy to plot the anisotropic range.

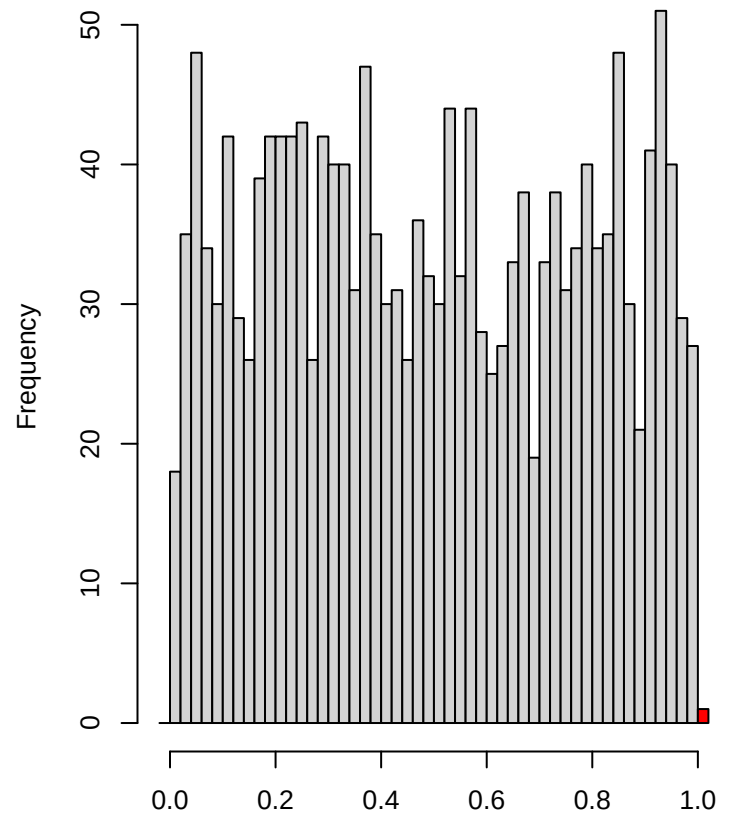
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy	14	4806.972	5.175365
spatiotemporal_random_fields_ar1	12	4812.148	32.354809
spatiotemporal_random_fields_iid_with_anisotropy	13	4844.503	1.712003
spatiotemporal_random_fields_iid	11	4846.215	9.707724
spatial_random_fields	11	4855.922	2.184446
spatial_random_fields_with_anisotropy	13	4858.107	42.139376
error_only	9	4900.246	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



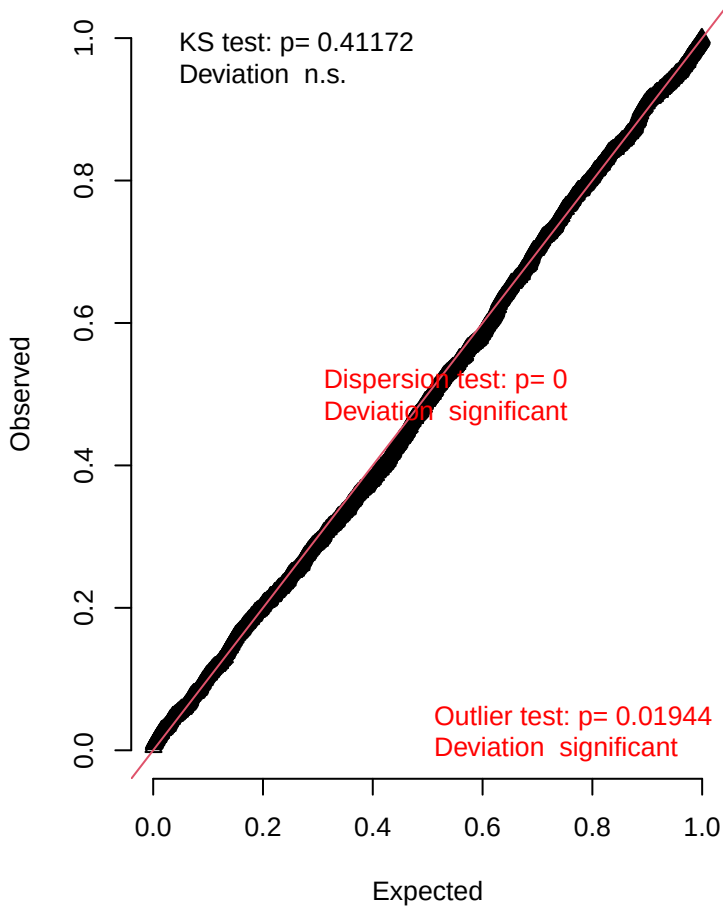
Outlier test significant



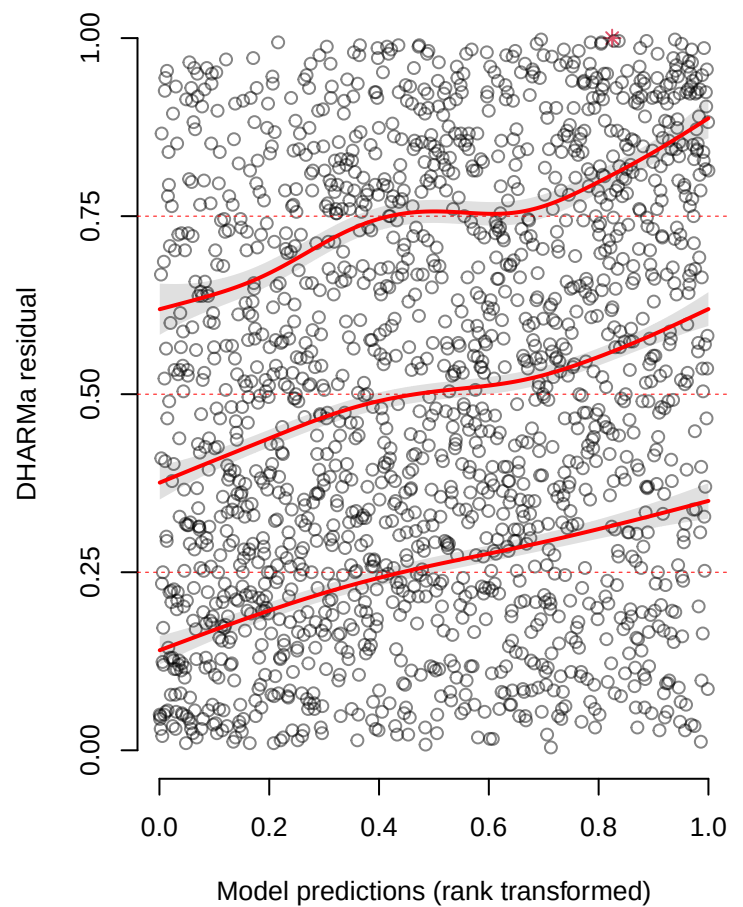
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals

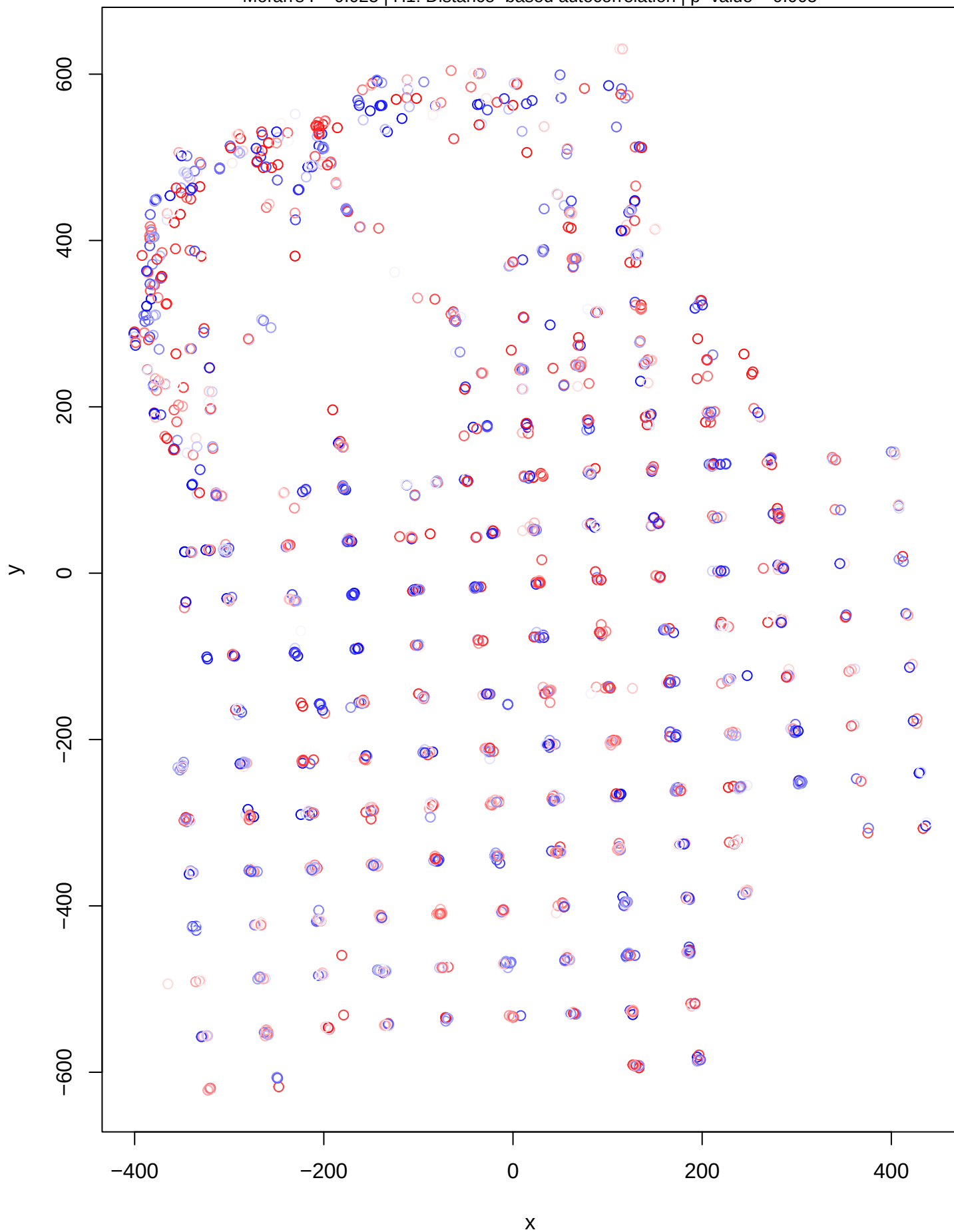


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant

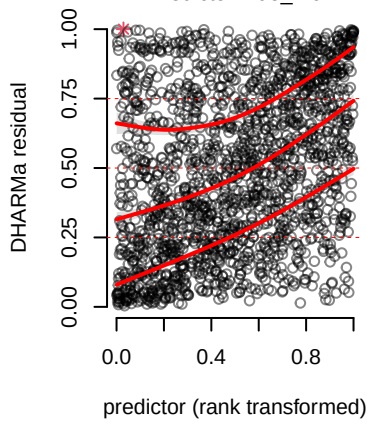


DHARMa Moran's I test for distance-based autocorrelation

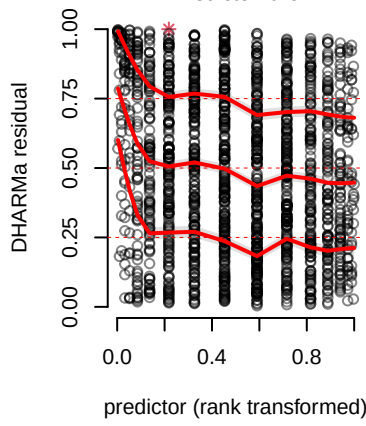
Moran's I = 0.025 | H1: Distance-based autocorrelation | p-value = 0.005



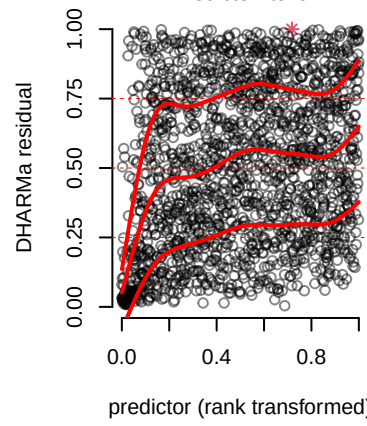
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



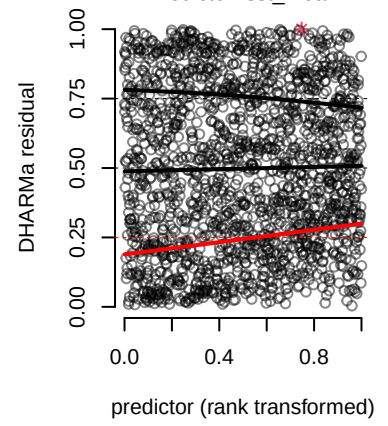
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



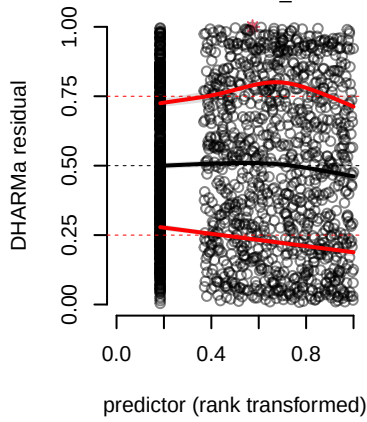
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



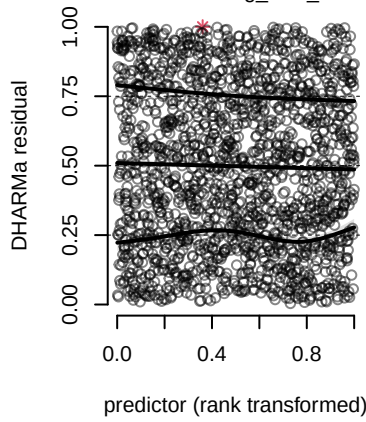
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



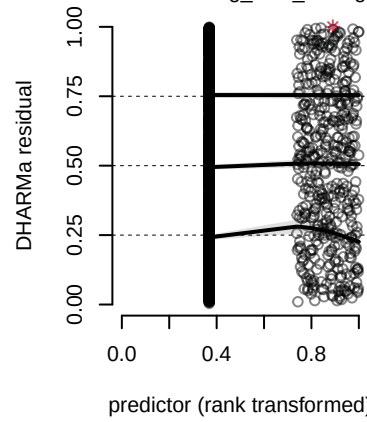
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



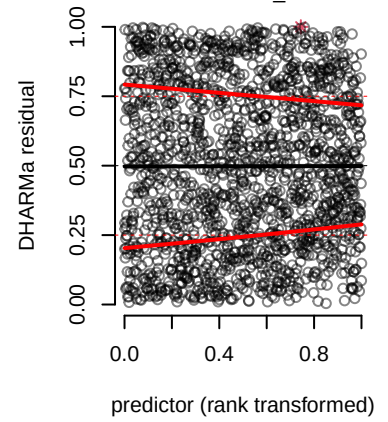
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_sd



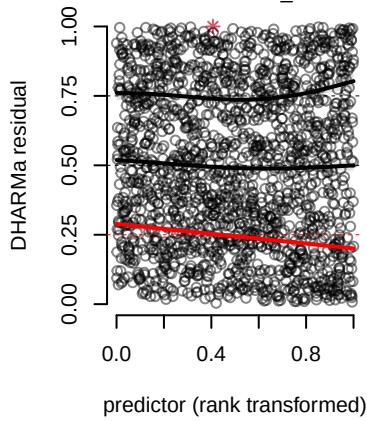
DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_sum_fishing



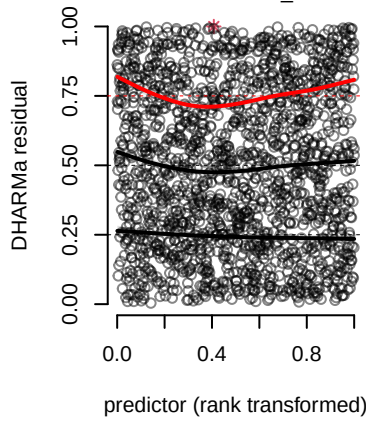
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



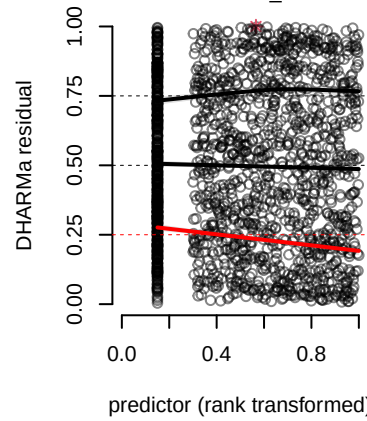
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



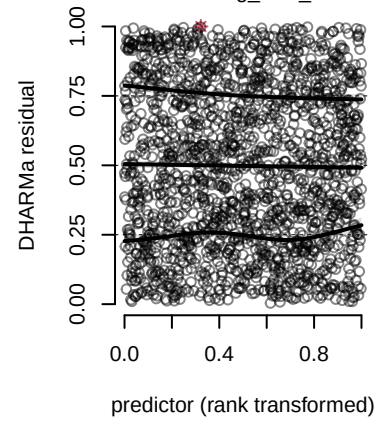
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_mean



DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant

